THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of triangles with whole-number sides (in cm) and perimeter 9 cm is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

2. The exterior angle at the base of an isosceles triangle is 110°. The apex angle is: 1

(a) 50° **(b)** 60° **(c)** 30° **(d)** 40°

3. In $\triangle ABC$, $\angle A = \angle B + \angle C$. Then $\angle A$ is: 1

(a) 45° **(b)** 90° **(c)** 60° **(d)** 120°

4. The angles of a triangle are $(2x)^\circ$, $(3x + 5)^\circ$ and $(4x - 50)^\circ$. The largest angle is: 1

(a) 80° **(b)** 100° **(c)** 70° **(d)** 90°

5. Sides 11 cm, 6 cm and 5 cm are given. Which statement is true? 1

(a) only a right triangle is possible **(b)** only an equilateral triangle is possible

(c) no triangle can be formed **(d)** a triangle can be formed

6. In an isosceles triangle with perimeter 34 cm the equal sides are 12 cm each. The base (cm) is: 1

(a) 9 **(b)** 12 **(c)** 11 **(d)** 10

7. The two acute angles of a right triangle are in the ratio $1 : 2$. The smaller acute angle is: 1

(a) 30° **(b)** 20° **(c)** 40° **(d)** 50°

8. The number of isosceles triangles with whole-number sides and perimeter 14 cm (equilateral included) is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 2

9. **Assertion (A):** A triangle cannot be formed with sides 3 cm, 4 cm and 8 cm. 1

Reason (R): The sum of any two sides of a triangle must be greater than the third side.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A triangle can have at most one right angle. 1

Reason (R): A scalene triangle has three unequal sides.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$ and $\triangle PQR$ it is known that $AB = PQ = 5$ cm, $BC = QR = 6$ cm and $\angle A = \angle P = 50^\circ$. Are the triangles necessarily congruent? Give the rule or the reason. 2

12. In isosceles $\triangle ABC$ with $AB = AC$, side BC is extended to D and $\angle ACD = 115^\circ$. Find $\angle A$ and $\angle B$. 2

13. List all isosceles triangles with whole-number sides and perimeter 20 cm (equal sides a, base b). How many are there? 2

14. In $\triangle ABC$ and $\triangle PQR$ it is known that $\angle A = \angle P = 40^\circ$, $\angle B = \angle Q = 65^\circ$ and $\angle C = \angle R = 75^\circ$. Are the triangles necessarily congruent? Give the rule or the reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, $\angle A = 50^\circ$. The exterior angles at B and C are $(4x + 60)^\circ$ and $(3x + 100)^\circ$. Find x and the angles B and C. 3

16. The perimeter of an isosceles triangle is 40 cm. Each equal side is 2 cm longer than the base. Find the sides and show that such a triangle exists. 3

17. Two sides of a triangle are 8 cm and 21 cm. The third side is a whole number of cm divisible by 5 . List all possible lengths and the largest perimeter. 3

OR

In $\triangle ABC$, $\angle A = 40^\circ$. The exterior angles at B and C are $(5x + 50)^\circ$ and $(4x + 62)^\circ$. Find x and the angles B and C. 3

18. The perimeter of an isosceles triangle is 45 cm. Each equal side is 3 cm longer than the base. Find the sides and show that such a triangle exists. 3

OR

Two sides of a triangle are 6 cm and 19 cm. The third side is a whole number of cm divisible by 4 . List all possible lengths and the largest perimeter. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Meena has sticks of various lengths and tries to make triangles. 5

(i) Can she form a triangle with sticks of 4 cm, 5 cm and 9 cm? Give the reason.

(ii) Can she form one with 5 cm, 6 cm and 10 cm?

(iii) Can she form one with 7 cm, 7 cm and 12 cm? What type of triangle is it?

(iv) Two sticks are 6 cm and 9 cm. Between which two lengths must the third stick lie, and how many whole-number lengths are possible?

20. The exterior angles of a triangle (one at each vertex) are in the ratio $4 : 5 : 6$. Find the three exterior angles, the three interior angles and classify the triangle. 5

OR

$\triangle ABC \cong \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Given $\angle A = (3x + 8)^\circ$, $\angle D = (5x - 14)^\circ$, $AB = (2y + 5)$ cm and $DE = (3y - 3)$ cm. Find x, y, $\angle A$ and AB. If $\angle B = 30^\circ$, find $\angle F$. 5

END OF QUESTION PAPER — Total: 40 Marks